Quark Matter Phase Boundary Evolution in Neutron Star Mergers

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Place holder

I. INTRODUCTION

[MGA I plan to do significant editing of the Intro once the rest of the paper is in good shape. Until then let's leave it as it is.] Neutron star mergers provide a promising opportunity to study the dynamics of neutron star matter on the millisecond timescale. It is hoped that this information will help us to map the phases of ultra-dense matter. In particular, if there were a first-order phase transition from nuclear matter to quark matter at some critical pressure, it could produce signatures in the gravitational or electromagnetic waves emitted, either during the inspiral or after the merger. AH cite both PRLs already here? In both scenarios, the speed at which the boundary moves is an essential factor which in current treatments is treated by extreme approximations, such as setting it to zero or infinity, as we describe below. In this work we estimate how fast the boundary is likely to propagate under conditions typically found during the inspiral and the merger phase, and draw conclusions about the validity of those approximations.

During inspiral, a phase boundary between nuclear and quark matter may support interface modes whose excitation produce signatures that are potentially detectable with the current LVK network, and certainly within reach of next-generation detectors like Cosmic Explorer or the Einstein Telescope [1, 2]. The existence and frequency of the interface mode critically depend on the conversion timescale, i.e., the speed with which the phase boundary between nuclear and quark matter can move via conversion of metastable nuclear matter to quark matter, or vice versa. If the phase boundary can move too fast, the restoring force for the interface mode and thus the mode itself vanish. All recent analyses [1–3] set this velocity to zero and assume that on the millisecond timescale of the oscillation, the phase content is frozen.

[MGA Broaden this part to include nuclear/quark transitions in supernovas? e.g. https://arxiv.org/abs/2108.00196 https://arxiv.org/abs/2603.18153] The same phase transition may also power core-collapse supernovae: if the collapsing core of a massive star reaches the density where nuclear matter converts to quark matter, the resulting quark core can trigger a delayed second shock and explode stars that neutrino heating alone fails to explode [4–6]. This scenario has a gravitational-wave signature of its own: the formation of a quark matter core inside a newly born proto-neutron star has been proposed as a source of gravitational waves in the MHz band, a supernova analog of the interface mode discussed above for the inspiral [7]. The boundary speed calculated here bears on both the explosion mechanism and this gravitational-wave signature, though core collapse reaches densities and temperatures beyond the range we consider below. In the post-merger phase there are also possible signatures of a nuclear-quark phase boundary. Several recent works have addressed this question using model equations of state that include a nuclear/quark transition [8–17], looking for features of the gravitational wave emission arising from the redistribution of mass as regions of the denser phase grow or shrink in response to pressure gradients in the violent collision between the stars. Current simulations assume that if a region of nuclear matter is compressed to a pressure above the critical pressure $p_{\rm crit}$ where quark matter becomes favored, it will immediately convert to quark matter, so there will be no significant regions of metastable nuclear matter or out-of-equilibrium quark matter AH we need to verify this as Veronica's EOS that the Most PRL uses does have out-of-equilibrium QM. This assumes that the phase boundary can travel distances of order kilometers in milliseconds, i.e., at speeds of order $10^6\,\mathrm{ms}^{-1}$. Our results will show that this assumption is questionable. AH small sentence about DM induced conversion works, and glitch mechanism via mtastable phase as further science cases of relevance AH microscopic = strong interaction time scale? In both situations described above, the key question is how fast a region of metastable nuclear matter could be converted to quark matter via movement of an existing phase boundary arising from a gradual (compared to microscopic timescales) increase in pressure. The propagation of a nuclear/quark phase boundary has been analyzed before, mostly for the more extreme case of the conversion of hadronic stars into strange stars in the context of the strange matter hypothesis [MGA For now I'm not separating ones that allow for a pressure-induced transition] [18–28] This paper complements earlier work by studying a slowly varying pressure driven transition to quark matter (rather than the global instability assumed

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by the strange matter hypothesis), studying the temperature and density ranges typical of neutron star mergers and supernovas, giving a survey of how the boundary speed varies as a function of the density and temperature of the nuclear phase, and providing an approximate analytic expression for the boundary speed. To achieve this we adopt a simplified model, focusing on the dominant flavor equilibration channel in quark matter. We do not consider the reverse process, conversion of quark matter to nuclear matter, although this may also occur in situations where there are pressure oscillations. We aim to set an approximate upper bound on the propagation speed of the boundary, so our approximations and simplifications are chosen to favor fast propagation.

[AH]mention neutron matter here as approximation? Our treatment assumes a deflagration-type conversion process, in which the pressure has risen high enough so that nuclear matter becomes metastable relative to 3-flavor quark matter, so weak interactions are required for the phase boundary to move. There is a more extreme possibility, only realized if the pressure becomes so high (e.g. in an externally induced shock front) that quark matter with the same flavor content as nuclear matter is energetically preferred. This opens up the possibility of a detonation-type conversion [25, 29] that does not require weak interactions and hence can proceed much faster. Relativistic jump-condition analyses classify detonation by a supersonic upstream velocity and typically find front speeds of order 0.1 times the speed of light or greater [30]. Such speeds are many orders of magnitude above our phase boundary velocity results and are therefore inaccessible when conversion is limited by weak-interaction-driven flavor equilibration. Detonation would require a separate shock-driven mechanism outside the regime considered in this work. [AH]this paragraph seems somehwat duplicated, big jump in technicality, and takes away results

A more violent version of this process, the quark-nova, is the hypothesized runaway conversion of an entire neutron star into a quark star, invoked as an energy source for superluminous supernovae and other high-energy transients [22, 28]. The microphysical boundary speed derived here is the starting point for such global conversion scenarios; extending our steady, planar treatment to the turbulent, whole-star regime of a quark-nova is left for future work. [AH]also duplicate content partially In Sec. II, we present our assumptions and the steady hydrodynamic equations. Sections III and IV describe the isothermal and adiabatic calculations of the boundary speed, respectively. In Sec. V we present our results, and in Sec. VI we summarize our conclusions and describe directions for future work. Our calculations use natural units, $\hbar = c = k_B = 1$ and metric signature $(-+++)$.

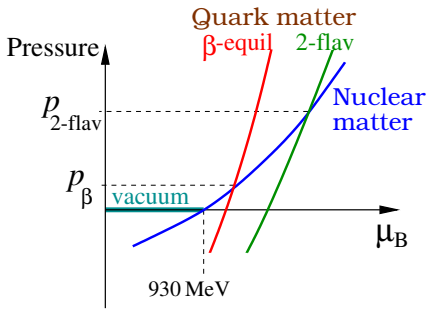


FIG. 1: Pressure comparison between phases at some arbitrary temperature. Above pressure P_β, 3-flavor chemically equilibrated quark matter is favored over nuclear matter. At $P > P_{2\text{-flav}}$, quark matter with the same flavor content as nuclear matter becomes favored, so weak interactions are no longer required for the conversion of nuclear to quark matter.

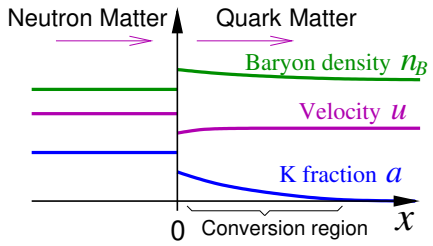


FIG. 2: Schematic illustration of the expected boundary profile. Note the discontinuities in some thermodynamic properties at the phase boundary located at $x = 0$, followed by relaxation to beta equilibrium as $x \to \infty$. In the adiabatic case, temperature will also show a discontinuity at $x = 0$ followed by relaxation to an asymptotic value as $x \to \infty$.

II. FORMALISM AND ASSUMPTIONS

A. Main assumptions

We assume the phase boundary is well approximated as a flat plane which can therefore be modeled as a one-dimensional system. This is a reasonable approximation for neutron star mergers, where the curvature of the boundary is on the kilometer scale, while its thickness is of order micrometers (see Eq. (45)). We defer any discussion of instabilities that would lead to a more complicated front geometry. [AH]As far as I know, people assume that instabilities work on the cm scale, which would still give us many orders of magnitude compared to micrometer. So the planar assumption should be fairly solid We assume that the surface tension of the phase boundary is large enough that mixed phases are not favored. We model the nuclear phase as pure neutron matter described by the QMC-RMF3 mean field theory [31–33]. We model quark matter as a Fermi gas of massless noninteracting up, down, and strange quarks [34, 35].

We will assume that the phase boundary is at a pressure where, in the range of temperatures relevant to neutron stars, 3-flavor beta-equilibrated quark matter is favored over nuclear matter (approximated as neutron matter in our case), but 2-flavor quark matter with the same flavor content as nuclear matter is not favored. I.e., the ambient pressure P is between P_β and $P_{\text{2-flav}}$ in Fig. 1. Thus weak interactions are required to complete the conversion of the metastable nuclear matter to quark matter. (In the adiabatic case the comparison is more complicated because the two phases are at different temperatures, see Sec. IV). We will follow the standard approach to nuclear to quark conversion [18, 26, 27, 30, 36, 37] which exploits the clear separation between strong-interaction-driven processes such as local thermal equilibration and deconfinement (which can be treated as instantaneous on the time scales relevant to neutron star mergers) and flavor equilibration, which occurs via weak interactions and is therefore much slower and must be treated explicitly.

We focus on the fastest flavor equilibration channel, the nonleptonic process

$$d\ u \leftrightarrow u\ s\ . \tag{1}$$

This equilibrates the quantum number that we will call "K", which measures the imbalance between down and strange quarks. We ignore electromagnetism, electrons, and neutrinos. We therefore neglect Urca processes, which are generally slower than the non-leptonic process [38–40], so their inclusion will make the boundary speed even lower.

As noted above, we assume local thermal equilibration is fast enough so that each fluid element has a well-defined temperature T. We will first write down the hydrodynamic equations that describe the movement of the fluid, including radiative loss and diffusion of flavor (the K quantum number) and heat. In the body of the paper, however, we will not explicitly treat the radiation or diffusion of heat via thermal conduction. Instead, we will calculate the boundary propagation speed for two extreme cases: isothermal (the whole system is thermally equilibrated to the same temperature) and adiabatic (no thermal conduction). Moreover, as our calculation including relativistic correction shows, the phase boundary doesn't move relativistically fast for wide range of temperature and baryon number densities, see Section V, so in both isothermal and adiabatic cases, slow velocity approximation is used. [JS Might need another appendix showing relativistic v.s. non-relativistic]

B. Formalism

To calculate the speed at which the nuclear/quark phase boundary propagates through neutron matter, we will look for a steady state solution to the equations that describe the propagation of a deflagration front.

We work in the rest frame of the phase boundary, which is located at $x = 0$, as illustrated schematically in Fig. 2. At $x < 0$ there is pure neutron matter which we take to be homogeneous, neglecting any disturbance propagating ahead of the phase boundary. The neutron matter flows towards the boundary with a velocity $u(x = 0^-)$, which we aim to calculate. Matter at $x = 0^-$ immediately deconfines into denser quark matter at $x = 0^+$ with enough strangeness to be created from neutron matter under the prevailing conditions [MGA These weasel words are meant to cover the isothermal vs adiabatic vs everything in between]. AH I don't understand the enough strangeness part? [MGA If the quark matter doesn't have enough strangeness then it won't be energetically favored (extreme case: udd matter).]

At $x > 0$ there is a conversion region where strange quarks are created via the weak interaction, leading to a flavor gradient that drives diffusion of strange quarks upstream towards the boundary. This is how the freshly created quark matter at $x = 0^+$ can immediately have a larger strangeness fraction than was present in the incoming neutron matter. As $x \to \infty$ the quark matter approaches a fully beta-equilibrated state.

The parameters that describe the state of the matter at any given position x are:

$$\begin{array}{ll} n_B & \text{baryon number density} \\ n_K \equiv \dfrac{n_d - n_s}{2} & \text{"K" density} \\ u & \text{velocity in the } x \text{ direction} \\ T & \text{temperature} \end{array} \tag{2}$$

From these one can, via the nuclear and quark equations of state (see Appendix A), calculate other properties such as pressure P, baryon chemical potential μ_B, energy density, and the K chemical potential

$$\mu_K = \mu_d - \mu_s\ , \tag{3}$$

which relaxes towards zero via the weak interaction (1).

We will find it useful to define the local K fraction

$$a(x) \equiv \frac{n_K(x)}{n_B(x)}\ . \tag{4}$$

C. Hydrodynamic equations

Although our calculations will focus on the isothermal and adiabatic regimes, we first present the more general hydrodynamic equations including the effects of flavor and heat diffusion and radiative energy loss but neglecting viscous stresses.

The stress-energy tensor without viscosity terms takes the form [41, 42]

$$T^{\mu\nu} = (P + \epsilon)U^\mu U^\nu + P g^{\mu\nu} + U^\mu q^\nu_{\text{th}} + q^\mu_{\text{th}} U^\nu\ , \tag{5}$$

and the fluid flow is defined by the baryon current

$$J^\mu_B = n_B U^\mu\ , \tag{6}$$

where $U^\mu = (\gamma, u, 0, 0)$ is the baryon fluid four-velocity in any frame in which the three velocity v in that frame is related with u and Lorentz factor γ by $u = \gamma v$ and $\gamma = 1/\sqrt{1-v^2}$, and $q^\nu_{\rm th}$ is the heat current which moves independently from the baryon current, i.e., $U_\mu q^\mu_{\rm th} = 0$.[LB?]

Baryon number is conserved, while radiation can carry energy and momentum out of the fluid:

$$\begin{aligned} \partial_\mu J^\mu_B &= 0\,, \\ \partial_\mu T^{\mu\nu} &= -Q^\nu\,. \end{aligned} \tag{7}$$

Here Q^ν is the radiation energy-momentum loss per unit spacetime volume. There is also a K quantum number current, whose conservation is violated by weak interactions,

$$\partial_\mu J^\mu_K = -\Gamma_K\,, \tag{8}$$

where Γ_K is the rate of down-to-strange quark conversion (see Appendix B) and

$$J^\mu_K = n_K U^\mu + D^\mu_K\,, \tag{9}$$

where D^μ_K is the strong-interaction-driven K diffusion current

$$D^\mu_K = -D_K(g^{\mu\nu}+U^\mu U^\nu)\partial_\nu n_K\,. \tag{10}$$

This definition ensures that there is no time component of the diffusion current in the local fluid rest frame, i.e., $U_\mu D^\mu_K = 0$.

[LB My goal was that up to this point, the focus is on which quantities are and are not conserved.]

For a steady state in the 1-dimensional phase boundary rest frame, derivatives with respect to t, y, and z vanish. Then baryon number conservation gives

$$\frac{d}{dx}(n_B u) = 0\,, \tag{11}$$

and energy-momentum conservation reduces to [LB Check the minus signs.][JS We using -+++ metric convention so I think it is plus $+P$]

$$\frac{d}{dx}\left[(P+\epsilon)\gamma u + \frac{\gamma^2+u^2}{\gamma} q^x_{\rm th}\right] = -Q^0\,, \tag{12}$$

$$\frac{d}{dx}\left[(P+\epsilon)u^2 + P + 2u q^x_{\rm th}\right] = -Q^x\,. \tag{13}$$

Note that we have used $q_\mu U^\mu = 0$ and $\gamma^2 - u^2 = 1$.

For the heat current, we model $q^\mu_{\rm th}$ using Eckart heat law [41, 43]

$$q^x_{\rm th} = -\kappa_{\rm th}\gamma^2 \frac{dT}{dx}\,, \tag{14}$$

where $\kappa_{\rm th}$ is the thermal conductivity [44], see Appendix D for more details.

In our 1-dimensional geometry we write the spatial component of the K current as

$$J_K = u n_K - \gamma^2 D_K \frac{d}{dx} n_K\,. \tag{15}$$

See Appendix C for our approach to estimating the diffusion coefficient D_K.

Neglecting radiative effects $Q^\mu = 0$ and expanding terms to $\mathcal{O}(u^2)$[LB Let us either write it fully relativistic or take the slow velocity limit now, to avoid issues with eq 16c], we arrive at the following set of hydrodynamic equations

$$\frac{d}{dx}(u n_B) = 0\,, \tag{16a}$$

$$\frac{d}{dx}\left[(P+\epsilon)u - \kappa_{\rm th}\left(1+\frac{5}{2}u^2\right)\frac{dT}{dx}\right] = 0\,, \tag{16b}$$

$$\frac{d}{dx}\left[P + (P+\epsilon)u^2 - 2u\kappa_{\rm th}\frac{dT}{dx}\right] = 0\,, \tag{16c}$$

$$\frac{d}{dx} J_K = -\Gamma_K\,. \tag{16d}$$

In the following sections we will solve Eqns. 16 to obtain the propagation speed of the phase boundary u in the isothermal and adiabatic limits.

The isothermal limit (Sec. III) is obtained by assuming infinite thermal conductivity, so the system will have a uniform temperature.

The adiabatic limit Sec. IV) is obtained by setting the thermal conductivity to zero, so entropy is conserved, apart from contributions from the weak interactions that relax the system towards beta-equilibrium[LB check this]. The steady-state boundary profile will then have a spatially varying temperature.

Anticipating the results that will be presented in Sec. V, which find that the typical boundary propagation speeds are around a millionth of the speed of light, we will drop relativistic corrections from now on.

III. BOUNDARY PROPAGATION SPEED: ISOTHERMAL CASE

A. Isothermal hydrodynamic equations

[LB Jan: we can mention that constant pressure is a good approximation but we should first write down the equations in the isothermal limit. It is not clear how Eq 16c survives the isothermal limit... unless taking the low velocity limit first is necessary.] [MGA We should start here.]

The isothermal limit is obtained by taking $\kappa_{\rm th} \to \infty$ in Eqns. 16. This yields

$$\frac{d}{dx}(un_B) = 0\,, \tag{17a}$$
$$\frac{d}{dx}T(x) = 0\,, \tag{17b}$$
$$\frac{d}{dx}P(x) = 0\,, \tag{17c}$$
$$\frac{d}{dx}J_K = -\Gamma_K\ , \tag{17d}$$

where we obtained Eq. (17b) from the $\kappa_{\rm th} \to \infty$ limit of Eqs. (16b) (or (16c)), and Eq. (17c) by eliminating $\kappa_{\rm th}$ between Eqs. (16b) and (16c) and discarding higher order velocity terms. The first three equations give us simple constancy conditions,

$$\left.\begin{aligned} J_B &= n_B u\,, \\ \Pi &= P\,, \\ T & \end{aligned}\right\} \text{independent of } x. \tag{18}$$

Thus to obtain the profile of the moving phase boundary we only need to solve the K flux equation Eq. (17d), a second-order differential equation that can be recast as two first-order differential equations for the K density and current,

$$\frac{d}{dx}n_K = \frac{un_K - J_K}{D_K} \tag{19a}$$
$$\frac{d}{dx}J_K = -\Gamma_K\,. \tag{19b}$$

where D_K and Γ_K in quark matter ($x > 0$) are given in Appendices C and B. We do not need to know D_K or Γ_K in nuclear matter ($x < 0$), because we assume it is uniform and equilibrated. At the boundary itself we expect Γ_K will remain finite, but the right side of Eq. (19a) will include a delta-function singularity (as the corresponding equation for dn_B/dx does, since u and n_B both have discontinuities).

There is now a well defined procedure for constructing a solution to the hydrodynamical equations. Assume we have solved Eq. (19) up to some position x, giving $n_K(x)$ and $J_K(x)$. We can combine this with the already-specified values of the constants (18) and the quark matter equation of state (Appendix A) at the prescribed temperature to solve for the remaining properties of the matter at x, such as $n_B(x)$, $u(x)$, $\mu_B(x)$, $\mu_K(x)$, etc. This allows us to calculate $D_K(x)$ and $\Gamma_K(x)$ (see Appendices B and C) and thereby obtain $n_K(x+\delta x)$ and $J_K(x+\delta x)$.

B. Boundary conditions

To construct the boundary profile we need 5 initial conditions specifying the state of the incoming nuclear matter: $n_B(0^-)$, $n_K(0^-)$, $dn_K/dx(0^-)$, $T(0^-)$, $u(0^-)$. However, since we assumed the incoming matter is neutron matter, for which $n_K = n_d/2$, we have

$$n_K(0^-) = n_B(0^-)\,, \tag{20}$$

and since we have assumed the neutron matter is homogeneous,

$$\frac{dn_K}{dx}(0^-) = 0\,, \tag{21}$$

so we only need to specify $n_B(0^-)$, $T(0^-)$, $u(0^-)$, which fixes the conserved fluxes J_B and Π for the whole solution.

The first step in solving the equations is to relate the K density and current at $x = 0^+$ to the initial conditions at $x = 0^-$, after which we can evolve the solution to $x = +\infty$ using (19).

Integrating Eq. (19b) across the sharp interface yields the continuity condition $J_K(0^+) = J_K(0^-)$, which can be re-expressed using Eq. (19a) as a hydrodynamic jump condition,

$$u(0^+)n_K(0^+) - \left[D_K\frac{dn_K}{dx}\right]_{0^+} = u(0^-)n_K(0^-)\,. \tag{22}$$

We do not need to know the K diffusion coefficient on the nuclear side because we have assumed there is no gradient there.

We cannot integrate Eq. (19a) to obtain an analogous relation between $n_K(0^+)$ and $n_K(0^-)$, because the right side of Eq. (19a) includes an unknown delta-function divergence at $x = 0$. We therefore have to treat $n_K(0^+)$ as an additional parameter of the system. In our calculations we will choose the value of $n_K(0^+)$ that maximizes the speed of the boundary.

Since the solution must yield equilibrated quark matter as $x \to \infty$, we also impose the boundary condition

$$\mu_K(\infty) = 0\ , \tag{23}$$

which for our model of quark matter is equivalent to $n_K(\infty) = 0$. This turns (19) into an eigenvalue problem, where for given $n_B(0^-)$, $T(0^-)$, and $n_K(0^+)$ there is only one value of the incoming neutron matter velocity $u(0^-)$ that yields a valid solution.

We can now obtain the boundary speed in terms of the density and temperature of the neutron matter and the K fraction in the freshly created quark matter. We do this by solving (19) as described in Sec. III A using the boundary conditions described above.

Our results presented in Sec. V will be obtained by solving the differential equations numerically. However, as we now describe, one can also formulate an approximate analytic solution, which is helpful for interpreting the results.

C. Analytic approximation

To obtain an analytic approximate solution for the phase boundary propagation speed, we note that in the conversion region, from $x = 0^+$ to ∞, as μ_K relaxes to zero, the constancy of the pressure and temperature (18) requires the quark matter to follow an isobar in the (μ_B, μ_K) plane as shown in Fig. 3.

From the bag model equation of state (A3), we see that $d\mu_B/d\mu_K|_P \sim \mu_K/\mu_B$ which is of order 1%, so along this isobar we expect μ_B (and consequently n_B, which is mainly determined by μ_B) will only vary by a small amount, as seen in Fig. 3. We therefore develop an analytic approach based on approximating μ_B and n_B (and therefore the velocity u, see (18)) as constant from $x = 0^+$ to ∞, in addition to pressure and temperature, writing the x-independent baryon density in the quark matter as

$$n_B^{(+)} \equiv n_B(x) \quad \forall x > 0\,. \tag{24}$$

We define λ_n (which will always be less than 1) as the ratio of neutron matter to quark matter baryon density,

$$\lambda_n \equiv \frac{n_B(0^-)}{n_B^{(+)}}\,, \tag{25}$$

so the x-independent velocity of the quark matter is

$$u^{(+)} = \lambda_n u(0^-)\,. \tag{26}$$

Since n_B is independent of x, it is convenient to work in terms of the local K fraction $a(x)$ defined in Eq. (4).

To find the speed $u(0^-)$ we use Eq. (22) along with (24), (25), (26) to obtain

$$\begin{aligned} u(0^-) &= -\frac{D_K(0^+)a'(0^+)}{\lambda_n\big(1-a(0^+)\big)}\,, \\ \text{where}\quad a'(x) &\equiv \frac{da}{dx} = \frac{1}{n_B^{(+)}}\frac{dn_K}{dx}\,. \end{aligned} \tag{27}$$

The boundary conditions determine λ_n and $a(0^+)$, but $D_K(0^+)a'(0^+)$ is determined by the requirement that $a(\infty) = 0$. To impose this exactly we would have to evolve $a(x)$ from $x = 0^+$ to infinity using the K current equation Eq. (19).

We cannot solve this equation exactly in full generality, but we can make a good estimate of the result. Multiplying Eq. (19b) by $D_K(x)a'(x)$, using Eq. (19a) to eliminate J_K, and integrating from $x = 0^+$ to infinity, we find

$$\frac{1}{2}\Big(D_K(0^+)\,a'(0^+)\Big)^2 = -u^{(+)}I_1 + I_2\,, \tag{28}$$

where I_1 and I_2 are positive dimensionless integrals over x which can be rewritten as integrals over a, since $a(x)$ is expected to be a monotonically decreasing function over $0^+ \leqslant x < \infty$ as the quark matter equilibrates in the conversion region behind the boundary.

$$\begin{aligned} I_1 &= \int_{0^+}^{\infty} D_K(x)\big(a'(x)\big)^2\,dx\,, \\ &= -\int_0^{a(0^+)} D_K a'\,da\,. \end{aligned} \tag{29}$$

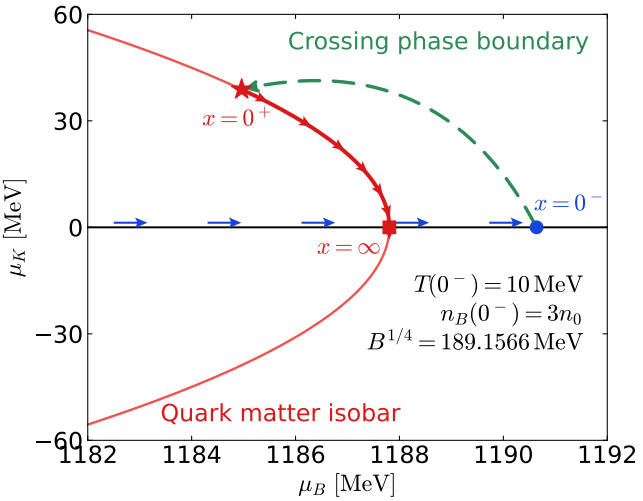


FIG. 3: Constant-pressure trajectory through the quark conversion layer in the (μ_B, μ_K) plane at fixed $T = 10\,\mathrm{MeV}$. The red curve is the quark-matter isobar for $n_B(0^-) = 3n_0$ and $B^{1/4} = 189.1566\,\mathrm{MeV}$. The red arrows run from the just-deconfined state at $x = 0^+$ to the equilibrated state at $x \to \infty$; the green arrow represents deconfinement from the nuclear-side marker at $x = 0^-$. [LB Can we make the x=0- to 0+ curve a straight line. It seems too suggestive that this is a particular path. Also are the blue arrows necessary?]

$$\begin{aligned} I_2 &= -\frac{1}{n_B^{(+)}}\int_{0^+}^{\infty} D_K\Gamma_K a'\,dx\,, \\ &= \frac{1}{n_B^{(+)}}\int_0^{a(0^+)} D_K\Gamma_K\,da\,. \end{aligned} \tag{30}$$

These expressions are valid however strongly D_K and Γ_K depend on x, so we will be able to use them in the adiabatic case also.

In an exact calculation both integrands should be obtained from the solution to Eq. (19), but we will now show how one can obtain reasonable approximations to both integrals without knowing the solution.

We first estimate I_1 because the estimate will turn out to depend on our desired result $u(0^-)$, requiring regrouping of Eq. 27. We assume that, as we see in the numerical solutions, the actual profile has $D_K(x)a'(x)$ increasing monotonically from its value $-\lambda_n u(0^-)[1 - a(0^+)]$ at $x = 0^+$ (from Eq. (27)) to zero at $x = \infty$. This is expected because $a(x)$ decreases monotonically to zero, and any variation in the diffusion constant D_K, which depends mainly on the temperature, is too weak to overcome that. We can then parameterize the integral in terms of a "sharpness" factor ξ:

$$\begin{aligned} I_1 &= -\frac{1+\xi}{2}\,D_K(0^+)a'(0^+)\,a(0^+) \\ &= \frac{1+\xi}{2}\,\lambda_n u(0^-)\,[1-a(0^+)]\,a(0^+)\,. \end{aligned} \tag{31}$$

Here $-1 < \xi < 1$ characterizes the shape of the diffusion profile. For details of this sharpness factor ξ and derivation of Eq. (31), see Appendix E. Substituting this relation into (28) and (27) yields

$$u^2(0^-) = \frac{2I_2}{\lambda_n^2[1-a(0^+)][1+\xi a(0^+)]}\,, \tag{32}$$

We now evaluate I_2 in the isothermal regime. The K decay rate Γ_K is given by Eq. (B3), evaluated at the constant quark-side baryon chemical potential $\mu_B^{(+)}$. At fixed T and $\mu_B^{(+)}$, the coefficients D_K and γ_K are constant, and Eq. (30) reduces to [JS Resolved:] [MGA Should sort this out in the appendix, giving a result in terms of μ_B]. [JS Resolved:] [MGA Do we need η? Why not write it explicitly each time.]

$$\begin{aligned} I_2 &= D_K\gamma_K \int_0^{a(0^+)} \left(a^3 + \frac{81\pi^2T^2}{(\mu_B^{(+)})^2}a\right) da \\ &= \frac{D_K\gamma_K\, a^2(0^+)}{4}\left[a^2(0^+) + \frac{162\pi^2T^2}{(\mu_B^{(+)})^2}\right]. \end{aligned} \tag{33}$$

Substituting Eq. (33) into Eq. (32) we obtain the analytic estimate of the phase boundary speed,

$$u_{\rm iso}^2(0^-) = \frac{D_K\gamma_K\, a^2(0^+)\big[a^2(0^+) + 162\pi^2T^2/(\mu_B^{(+)})^2\big]}{2\lambda_n^2\big(1-a(0^+)\big)\big(1+\xi a(0^+)\big)}\,, \tag{34}$$

where λ_n and ξ are defined in Eqs. (25) and (31), respectively.

The two terms shown in square brackets in the numerator arise from the cubic and linear contributions to the K decay rate in Eq. (B3). The cubic contribution dominates when $a(0^+) \gg 9\pi T/\mu_B^{(+)}$, while the thermal contribution dominates at higher temperatures or smaller $a(0^+)$.

The final step is to choose a value for $a(0^+)$, the K fraction in freshly created quark matter, which lies between 0 (corresponding to quark matter with as many strange quarks as there are down quarks), and 1 (corresponding to quark matter with no strangeness, i.e., having the same flavor content as neutron matter, $a(0^-) = 1$). We noted in Sec. III B that this is an unspecified parameter in our calculation. We will now choose it so as to give the maximum speed of the boundary $u_{\rm iso}^{\rm max}(0^-)$. Since the speed is a monotonically increasing function of $a(0^+)$ in the domain $0 < a(0^+) < 1$ for any value of the sharpness factor $-1 < \xi < 1$, this means we set $a(0^+)$ to $A_{\rm iso}$, the largest physically possible value of $a(0^+)$, which is given by the requirement that the quark matter at $x = 0^+$ must be thermodynamically favored over the neutron matter at $x = 0^-$. Since these two forms of matter are at the same pressure and temperature, neutron→ quark conversion is favored when the average chemical potential of a quark at $x = 0^+$ is lower than the average chemical potential of a quark in neutron matter at $x = 0^-$. Thus $A_{\rm iso}$ is specified as the value of $a(0^+)$ for which the average chemical potentials are equal. Quark matter with $a(0^+) > A_{\rm iso}$ would have insufficient strangeness to make it favored over neutron matter (see Fig. 1 and Ref. [24]).

Since Eq. (34) has monotonic dependence on $a(0^+)$ and $0 < a(0^+) < A_{\rm iso} < 1$, setting $a(0^+) = A_{\rm iso}$ and inserting the leading term of Eq. (C1) gives

$$\begin{aligned} u_{\rm iso}^{\rm max}(0^-) \simeq{}& (98\,{\rm m/s})\frac{n_B^{(+)}}{n_B(0^-)} \\ &\times \left(\frac{\mu_B^{(+)}}{1200\ {\rm MeV}}\right)^{\frac{17}{6}} \left(\frac{10\ {\rm MeV}}{T}\right)^{\frac{5}{6}} \\ &\times \frac{A_{\rm iso}\sqrt{A_{\rm iso}^2 + 162\pi^2T^2/(\mu_B^{(+)})^2}}{\alpha_s^{5/6}\sqrt{1-A_{\rm iso}}\sqrt{1+\xi A_{\rm iso}}}\,. \end{aligned} \tag{35}$$

[MGA Since we aren't using the explicit form of $A_{\rm iso}$, we could have given this equation earlier.][JS Done]

Writing the average chemical potential as $\mu_B(0^+) + n_K(0^+)\mu_K(0^+)/n_B(0^+)$ and Taylor-expanding the quark matter EoS around its beta-equilibrated state,[MGA Didn't we already do that in Appendix A? Can't we just cite an equation from there?] we can obtain an approximate formula [MGA Are there additional approximations beyond the Taylor expansion?] that in the parameter range that we explored agrees with the numerical result to within 1%,

$$A_{\rm iso} \approx \sqrt{\frac{2\big(\mu_B(0^-) - \mu_B(\infty)\big)}{n_B(\infty)}\left.\frac{\partial^2 P_{\rm QM}}{\partial\mu_K^2}\right|_\infty}\,. \tag{36}$$

IV. BOUNDARY PROPAGATION SPEED: ADIABATIC CASE

A. Adiabatic hydrodynamic equations

The adiabatic limit is obtained by setting thermal conductivity $\kappa_{\rm th} = 0$ and radiative energy loss $Q_\nu = 0$ in Eq. (16). This gives the adiabatic steady-state equations,

$$\frac{d}{dx}(un_B) = 0\,, \tag{37a}$$

$$\frac{d}{dx}((P+\epsilon)u) = 0\,, \tag{37b}$$

$$\frac{d}{dx}((P+\epsilon)u^2 + P) = 0\,, \tag{37c}$$

$$\frac{d}{dx}J_K = -\Gamma_K\,. \tag{37d}$$

The baryon and momentum fluxes remain those in Eq. (18), while Eq. (37b) fixes the advective energy flux,

$$E = hu\,. \tag{38}$$

The upstream values $n_B(0^-)$, $T(0^-)$, and $u(0^-)$ fix E for the whole solution. We solve the same K transport equations (19), with the boundary conditions of Sec. III B.

The reconstruction described in Sec. III A now also determines $T(x)$from energy conservation, rather than prescribing it. Most of our results in Sec. V were obtained by solving these adiabatic equations numerically.

B. Analytical approximation

We use the piecewise constant approximation of Sec. III C, now retaining the x dependence of temperature $T(x)$ rather than fixing it. As we will now show, this allows us to find an analytic expression for the speed of the boundary.

The density and velocity definitions in Eqs. (24)–(26), the interface gradient in Eq. (27), and the general speed relation in Eq. (32) apply here as well.

To evaluate I_2, we need to know how the K diffusion coefficient D_K and K creation rate Γ_K vary through the conversion region. In the piecewise constant approximation the pressure and baryon chemical potential have constant values $P^{(+)} \approx \Pi$ and $\mu_B^{(+)}$ in that region, so the system follows an isobar in the μ_K-T plane. Moreover, since $\mu_K/\mu_B^{(+)}$ is typically of order 0.1 and $T/\mu_B^{(+)}$ is of order 0.03, we can expand the quark matter EoS to quadratic order (A3), so the isobar takes the form of a quadrant of an ellipse in the μ_K-T plane. To this order there is a linear relationship between the K fraction a and μ_K (Eq. (A6)) so the quark matter follows an elliptical quadrant in the a-T plane, as shown in Fig. 4.

Using Fig. 4 we can describe the conversion region as follows. At $x = 0^+$ the quark matter has some initial K fraction $a(0^+)$ which determines the initial temperature $T(0^+)$ (the red star in the figure). As x increases, the system moves upwards along the elliptical isobar, ending at $a = 0$ and some final temperature when $x \to \infty$ (the red square in the figure).

This sets a limit on the previously unconstrained parameter $n_K(0^+)$, or equivalently $a(0^+)$. For given initial conditions $(n_B(0^-), T(0^-), u(0^-))$, the largest possible value for $a(0^+)$ is the one that would give the lowest allowed initial quark matter temperature, $T(0^+) = 0$. Thus, $a(0^+)$ cannot be larger than A [MGA I would like to have a more obvious symbol for this, but not sure what would be better], the value where the isobar meets the $T = 0$ axis in Fig. 4, defined by $P_{\rm QM}(A, T = 0) = P^{(+)}$, so to leading order,

$$a(0^+) \leqslant A \approx \frac{3}{2}\frac{\sqrt{108\pi^2\left(P^{(+)} + U_{\rm bag}\right) - \left(\mu_B^{(+)}\right)^4}}{\left(\mu_B^{(+)}\right)^2} \tag{39}$$

The isobar in Fig. 4 defines the quark matter temperature $T(x)$ as a function of $a(x)$; solving $P_{\rm QM}(a, T) = P_{\rm QM}(A, 0)$ we find

$$T^2 = \frac{2(\mu_B^{(+)})^2}{81\pi^2}(A^2 - a^2) \tag{40}$$

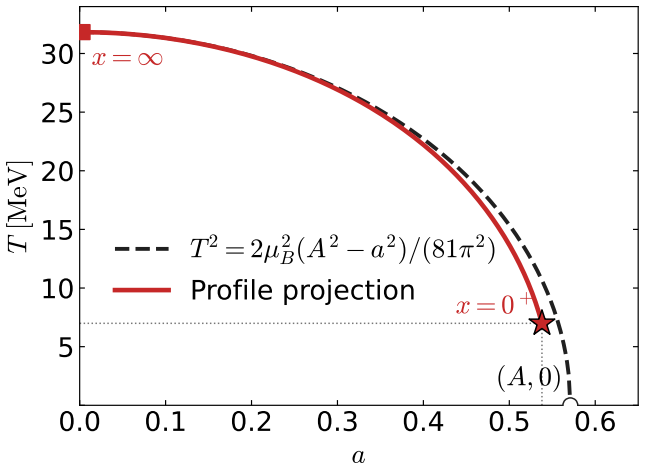


FIG. 4: Constant-pressure trajectory through the quark conversion layer. The red curve is the energy-conserving numerical profile for $T(0^-) = 10$ MeV, $B^{1/4} = 180$ MeV, $n_B(0^-) = 3n_0$, and $a(0^+) = 0.55$; the dashed curve is the quarter ellipse in Eq. (40). The profile runs from the just-deconfined state at $x = 0^+$ to the equilibrated state at $x \to \infty$. The point $(A, 0)$ is the zero-temperature endpoint of the extrapolated isobar and need not coincide with the state at $x = 0^+$. [LBWhy choose a+=0.55? Why not either choose a+ such that T+=T- (if this is possible) or choose a+ such that T-=0.1 MeV.]

We can now evaluate the integral I_2 (30) by using the results from Appendices B and C) for $\Gamma_K(a, T)$ and $D_K(a, T)$ (keeping only the first term), and using (40) to express T as a function of a. We find

$$I_2 = \frac{27\pi^{7/3}}{56\sqrt{2}}\frac{A^{7/3}\gamma_K}{\alpha_s^{5/3}h_D\mu_B^{(+)}} \times \left[8 + \left(1 - \frac{a(0^+)^2}{A^2}\right)^{1/6}\left(\frac{a(0^+)^2}{A^2} - 8\right)\right]. \tag{41}$$

Unlike the isothermal result in Eq. (33), here $D_K[T(a)]$ remains inside the integral and produces the $A^{7/3}$ prefactor and the fractional powers.

So from (32),

$$u_{\rm ad}^2(0^-) = \frac{9\pi^{7/3}}{4\sqrt{2}\,h_D\alpha_s^{5/3}}\frac{\gamma_K A^{7/3}}{\mu_B^{(+)}\lambda_n^2\left(1 - a(0^+)\right)\left(1 + \xi a(0^+)\right)} \times \left[\frac{24}{7} - 3\left[1 - \frac{a^2(0^+)}{A^2}\right]^{1/6} - \frac{3}{7}\left[1 - \frac{a^2(0^+)}{A^2}\right]^{7/6}\right]. \tag{42}$$

To find the fastest speed at which the phase boundary could move, $u_{\rm max}(0^-)$, we set $a(0^+)$, the K fraction in freshly created quark matter, to the value that maximizes the speed. Eq. (42) shows that the speed of the boundary grows as $a(0^+)$ gets larger. The factor in square brackets

has a weak dependence on $a(0^+)$, while the denominator $\big(1-a(0^+)\big)\big(1+\xi a(0^+)\big)$ decreases for every $-1<\xi<1$, as in Sec. III C. ~~For any reasonably smooth profile we expect $|\xi|\ll 1$~~ Comparing numerical profiles for different $T(0^-)$ and $n_B(0^-)$, we find $\xi=-0.5$ is a good approximation (Appendix E).

This means that to achieve the maximum phase boundary speed for given initial conditions, we need larger $a(0^+)$, which makes sense because that corresponds to a lower strangeness fraction in the freshly created quark matter, closer to the flavor content of the neutron matter. As noted above, the largest possible value of $a(0^+)$ is A, so setting $a(0^+)=A$ we find the maximum speed

$$u_{\rm ad}^{\rm max}(0^-) = (82\,{\rm ms}^{-1})\,\frac{n_B^{(+)}}{n_B(0^-)}\left(\frac{\mu_B^{(+)}}{1200\,{\rm MeV}}\right)^2\times \frac{A^{7/6}}{\alpha_s^{5/6}\sqrt{1-A}\sqrt{1+\xi A}}\,. \tag{43}$$

Thus the typical speed of the boundary is expected to be of order $100{\rm m\,s}^{-1}$. In this expression the only factor that could drive the speed to a significantly higher value is $1/\sqrt{1-A}$. To see when A could approach 1 we recall that A is the largest possible K fraction in the freshly created quark matter at $x=0^+$. The K fraction in neutron matter is 1, so if $A=1$ then we have such a high pressure that nuclear matter can convert to quark matter with the same flavor content while conserving energy. In this case no weak interactions are needed, so it is not surprising that our approach breaks down.

[JS move this to isothermal case or appendix] We can also use the piecewise constant approximation to estimate the thickness $L_{\rm conv}$ of the conversion region, which determines the range of curvature that is compatible with our assumption of a planar geometry. If the boundary moves slowly then when $a\ll 9\pi T/\mu_B^{(+)}$, which is guaranteed to occur at large x where $a\to 0$, the K current equation reduces to

$$\frac{d^2a}{dx^2} = \frac{\gamma_K}{D_K}\left(\frac{9\pi T}{\mu_B^{(+)}}\right)^2 a\,. \tag{44}$$

At large x the temperature reaches an asymptotic value $T(\infty)$, so we can estimate the thickness of the conversion region as

$$L_{\rm conv} \approx \frac{\mu_B^{(+)}}{9\pi T(\infty)}\sqrt{\frac{D_K}{\gamma_K}} = (0.2\,\mu{\rm m})\,\frac{1}{\alpha_s^{5/6}}\left(\frac{1200\,{\rm MeV}}{\mu_B^{(+)}}\right)^{7/6}\left(\frac{10\,{\rm MeV}}{T(\infty)}\right)^{11/6}, \tag{45}$$

where we evaluated D_K using (C1) with $T=T(\infty)$. and γ_K from Eq. (B3). This supports our use of the planar approximation, since the typical curvature of a quark/nuclear boundary in neutron stars will be of order kilometers, much larger than the thickness of the boundary.

[MGA Edited up to here]

V. RESULTS

- first, the set up

We discuss the energy conserving front system Eq. (37) first. We choose the bag constant by requiring cold neutron matter and cold equilibrated strange quark matter to coexist at neutron matter critical baryon number density $n_B^{\rm crit}(T{=}0)=3n_0$. This gives $U_{\rm bag}^{1/4}=189.2\,{\rm MeV}$. Since we are not including Urca processes, we set strange quark mass to zero. We also adopt the conventional phenomenological choice $\alpha_s=0.3$ for bag-model quark matter [35, 45]. For each pair of upstream variables $n_B(0^-)$ and $T(0^-)$, we evolve the K density and current, reconstruct the local fraction $a(x)=n_K(x)/n_B(x)$, and enforce the conserved baryon, momentum, and energy fluxes across the quark phase.

But to get a unique solution to Eq. (37), hence a unique value for $u(0^-)$, we still need a proscribed $a(0^+)$, or equivalently, $T(0^+)$. As mentioned in Sec. IV B, we can utilize this freedom to maximize the velocity by choosing $T(0^+)=0$, but practically that would make $D_K^{-1}=0$ hence giving us a non-Lipschitz initial condition for the ODE system Eq. (19), which gives us a family of possible solutions instead of one unique solution. So in practice, when using analytical approximation Eq. (32) or Eq. (42) we can set $a(0^+)=A$ or $T(0^+)=0$, but when numerically solving the full system Eq. (37) we need to set a small but non-zero $T(0^+)$ to avoid the non-Lipschitz initial condition. From our numerical results Fig. 8, we find that the velocity doesn't change over two times for $T(0^+)<1$ MeV, so we choose $T(0^+)=0.1$ MeV as a representative value for the numerical results.

A. Phase boundary speed

- about contour plot:
  - existence of stable neutron matter region, and why it takes that shape
  - $A>1$ region and why that shape
  - velocity in between, also mention it diverge as the input goes close to $A=1$

AH I think if we have Fig1 as schematic, we should have an equivalent figure here with an actual result?

VI. CONCLUSION

- Impact of instabilities, Urca processes, ... on phase boundary speed

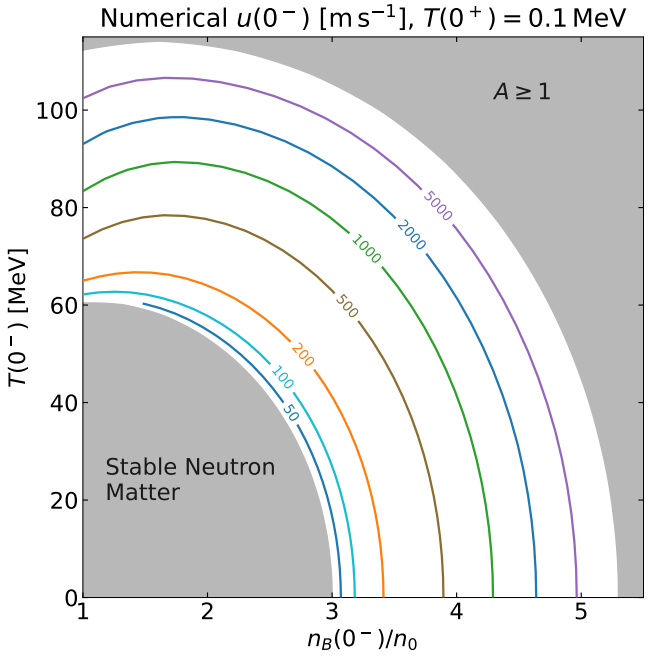


FIG. 5: Full numerical phase-boundary speed $u(0^-)$ in m s^{-1} as a function of the upstream baryon density and temperature, with $T(0^+) = 0.1$ MeV and $n_B^{\rm crit}(T{=}0) = 3n_0$. The white band is the domain in which the moving-front solution exists. Below the lower boundary neutron matter is stable against the assumed conversion, while the region marked $A \geq 1$ lies outside the weak-interaction-limited branch described by our model. The contours are drawn at 50, 100, 200, 500, 1000, 2000, and 5000 m s^{-1}.

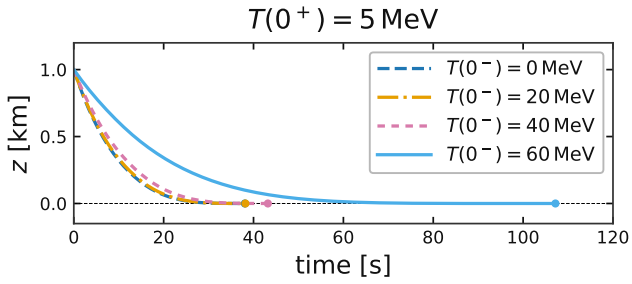


FIG. 6: Remaining distance $z(t)$ to the zero-speed surface, obtained from the full numerical speed map at fixed $T(0^+) = 5$ MeV. The initial distance is $z_0 = 1$ km, and the assumed density gradient is $0.3n_0$ km^{-1}. The points mark the fitted catch-up times for $T(0^-) = 0$, 20, 40, and 60 MeV. [LB I thought we were not including this figure?]

Our results allow us to speculate on the fate of interface modes in the inspiral. Typical frequencies of interface modes depend on the EoS of nuclear and quark matter as well as the star's mass, where heavier stars show lower frequency resonances. The frequencies themselves range from a couple hundred to around $f = 2000$ Hz (see Fig. 2 of Ref. [1]). Depending on the energy absorbed by the mode, the amplitude ranges from a few millimeters to centimeters in the inspiral, and might reach several meters in the postmerger. Assuming a frequency of $f = 700$ Hz, we can calculate the resonant velocity where the timescale of the oscillation matches the time given by the amplitude divided by the phase boundary velocity. In this resonant scenario, we expect viscous effects as described in [24] become important. At even larger velocities, we expect the mode to lose its support (see discussion in the nuclear case in Ref. [46]) and vanish. In the inspiral, the resonant velocity ranges from $v_{\rm res} = 1$ m/s to $v_{\rm res} = 15$ m/s, which is within the range found in our calculation shown in Fig. 5. In the postmerger scenario, the resonant velocity rises to $\mathcal{O}(1000\,{\rm m/s})$, which is reached in the strongly metastable scenario (see Fig. 5). This shows that any accurate calculation of the interface mode must include the finite phase boundary velocity and viscous effects, potentially via the dynamical sound-speed formalism developed in Ref. [46] and applied to quark matter in [cite Zhao Han Haber].

This also motivates a more detailed calculation of the phase boundary velocity:

- In this paper, we model the baryonic side as pure neutron matter. However, in compact stars, baryonic matter is composed of chemically equilibrated nuclear matter, which includes not only neutrons, but also protons, electrons, and muons in a composition determined by charge neutrality and chemical equilibration. Even hyperonic matter could be considered on the baryonic side of the phase boundary.

- The presence of leptonic particles opens up the possibility of proton or strangeness-changing Urca rates. The phase boundary velocity would then potentially be slowed down even further, as we now need to equilibrate multiple particle fractions on the quark matter side via the generally slower Urca processes. Our results thus likely present an upper bound on the velocity.

- At high temperatures, the physics of neutrinos might become important as neutrinos could cool down the conversion region, or if they get trapped, will alter the hydrodynamic description of the phase boundary. However, given our recent work in Refs. [47, 48], neutrino trapping on the microscopic scale of the phase boundary seems unfeasible, but could potentially play a role in the equilibrated baryonic or quark matter

- At the temperatures occurring in the inspiral, baryonic and quark matter might be found in a (color-)superconducting/superfluid state. Cooper pairing will not only alter the hydrodynamics of the problem, but suppresses the flavor-changing rates used in this work, and will also alter diffusive transport, thus furder slowing down the phase boundary.

- In this work, we have assumed that thermal equilibration across the phase boundary is happening on a timescale slower than the evolution of the phase boundary. Towards the end of the equilibration process, however, the phase boundary slows down ([AH]do we have a section explaining this?. To not only reach flavor, but also thermal equilibration, thermal equilibration across the boundary has to be taken into account. Otherwise, the final state of the evolution will be wrongly described as a state where the baryonic and the quark matter phase stay at different temperatures.
- In this work, we assume that all profiles on the nuclear matter side are constant, and neglect any disturbance propagating ahead of the phase boundary.
- Furthermore, the assumption of a steady state, which we deem reasonable, should be further investigated.
- Our hydrodynamic description is based on an ideal fluid description. Dissipative corrections might be taken into account in the stress-energy tensor, or be included via weak interactions as described in Ref. [49].
- If the phase boundary is driven by large-scale oscillations (as we expect it in the inspiral and the merger), the opposite scenario, where quark matter recombines to nuclear matter, has to be calculated. This will also allow us to estimate the strength of the so-called "phase conversion dissipation", a dissipative effect stemming from the out-of-phase movement of the actual phase boundary with the critical pressure [24].
- Cooper pairing: influence on diffusion, weak interaction rates, anything else?
- surface tension?

ACKNOWLEDGEMENTS

We thank XXX (Nils Andersson? Suprovo Ghosh?) for valuable feedback and discussions. MGA, LB, and AH are partly supported by the U.S. Department of Energy, Office of Science, Office of Nuclear Physics, under Award No. #DE-FG02-05ER41375. A.H. furthermore acknowledges financial support by the UKRI under the Horizon Europe Guarantee project EP/Z000939/1.

Appendix A: Bag model of quark matter

[MGA If we decide to impose electrical neutrality then this will need some rewriting.] In the approximation that we use for quark matter, ignoring electromagnetism and leptons, the pressure of quark matter is the sum of the pressures of the individual flavors,

$$\begin{aligned} P_{\rm QM}(\mu_B,\mu_K,T) &= P_q(\mu_B/3,T) + P_q(\mu_B/3+\mu_K/2,T) \\ &\quad + P_q(\mu_B/3-\mu_K/2,T) - U_{\rm bag}\,, \end{aligned} \tag{A1}$$

where μ_B is the baryon chemical potential and P_q is the pressure of a free Fermi gas of three colors of a single quark flavor. In this approximation the individual flavors have chemical potentials $\mu_u = \mu_B/3$, $\mu_d = \mu_B/3+\mu_K/2$, and $\mu_s = \mu_B/3-\mu_K/2$. We expect $\mu_K, T \ll \mu_B/3$ so we expand in powers of these to obtain

$$\begin{aligned} P_{\rm QM} = -U_{\rm bag} + \frac{\mu_B^4}{108\pi^2} + \frac{\mu_B^2}{12\pi^2}\mu_K^2 + \frac{\mu_K^4}{32\pi^2} \\ + \frac{\mu_B^2}{6}T^2 + \frac{1}{4}\mu_K^2 T^2 + \frac{19\pi^2}{36}T^4 + \cdots\,, \end{aligned} \tag{A2}$$

[MGA Do we use this 4th order form of the EoS anywhere?][JS No, we never use it again: only the quadratic truncation (A3) is cited later (line 480 and the analytic derivation in Sec. IV B). Worth keeping, though: these are exactly the terms we'd need if we later want to estimate how far the analytic formula should deviate from the full numerical result, since they're what the quadratic truncation drops. We don't have that comparison anywhere in Sec. V yet.] or to quadratic order,

$$P_{\rm QM} = -U_{\rm bag} + \frac{\mu_B^4}{108\pi^2} + \frac{\mu_B^2}{12\pi^2}\mu_K^2 + \frac{\mu_B^2}{6}T^2\,, \tag{A3}$$

which implies that to leading order,

$$n_K = \left.\frac{\partial P_{\rm QM}}{\partial \mu_K}\right|_{\mu_B,T} = \frac{\mu_B^2\mu_K}{6\pi^2}\,, \tag{A4}$$

$$n_B = \left.\frac{\partial P_{\rm QM}}{\partial \mu_B}\right|_{\mu_K,T} = \frac{\mu_B^3}{27\pi^2}\,. \tag{A5}$$

We can also express the quark matter properties in terms of the K fraction a. To this order,

$$a \equiv \frac{n_K}{n_B} = \frac{9\mu_K}{2\mu_B}\,. \tag{A6}$$

Appendix B: Strangeness creation

For positive μ_K, the nonleptonic process relaxes the K density at the rate [50]

$$\Gamma_K(\mu_B,\mu_K,T) = \mathcal{A}_K \mu_u^5 \mu_K \left(\mu_K^2 + 4\pi^2 T^2\right)\,, \tag{B1}$$

[JS u is a real participant, not an average: the reaction is $d+u \leftrightarrow u+s$ (Eq. (1)), with u a spectator on both sides, like the spectator nucleon in modified Urca. Madsen's rate has the same μ_u^5 prefactor for the same reason.][MGA In that case one would expect factors of μ_d and μ_s as well. I think Madsen is using μ_u as a symbol for something else, probably some average of chemical potentials. In our calculation everything should end up as a function of μ_B, μ_K, and T.] where

$$\mathcal{A}_K = \frac{16}{5\pi^5} G_F^2 \sin^2\theta_C \cos^2\theta_C\,, \tag{B2}$$

where $G_F = 1.16637 \times 10^{-11}\,\mathrm{MeV}^{-2}$ is the Fermi constant and $\theta_C = 13.02°$ is the Cabibbo angle. We use $\mu_u = \mu_B/3 + 2\mu_Q/3$, where electric charge chemical potential μ_Q is fixed by charge neutrality, $2n_u - n_d - n_s = 0$. [MGA But in the previous section we said $\mu_u = \mu_B/3$. Which are we actually using? Anyway, as noted above μ_u shouldn't be needed.] [JS Both, in different places: the exact rate used in the numerical code uses the full $\mu_u = \mu_B/3 + 2\mu_Q/3$ with μ_Q solved exactly from charge neutrality, while only the analytic approximation in Sec. IV B drops μ_Q and uses $\mu_u \simeq \mu_B/3$. Not an inconsistency, just two different calculations.] [MGA (1) In general we should show the same formulas that we use in the numerical calculation. If we are making some approximation beyond the ones we use in our numerics then we should say so explicitly. (2) In Appendix A we didn't impose electric neutrality. Our overall approach is to ignore electrons and electromagnetism, so it seems inconsistent to impose electric neutrality.]

For the analytic approximations in Secs. III C and IV B, we use the quadratic approximation of Sec. A to express the rate in terms of the K fraction. Using (A6) and $\mu_u \simeq \mu_B/3$, we obtain

$$\begin{aligned} \Gamma_K(a,T) &= n_B \gamma_K \left(a^3 + \frac{81\pi^2 T^2}{\mu_B^2} a \right) , \\ \gamma_K &= \frac{128}{3^8 \times 5\pi^3} G_F^2 \sin^2\theta_C \cos^2\theta_C \, \mu_B^5 . \end{aligned} \tag{B3}$$

Appendix C: Strangeness diffusion

The diffusion of the K flavor in quark matter is described by a diffusion coefficient D_K, which, since the quarks are relativistic, is closely related to the mean free path. This is determined by strong interactions, so we model it using the single gluon exchange perturbative expression [44, 51]

$$D_K^{-1} = \frac{8 N_f \alpha_s^2}{\pi} \left(h_D \frac{T^{5/3}}{q_D^{2/3}} + \frac{\pi^3 T^2}{12 q_D} \right) , \tag{C1}$$

Here $N_f = 3$ is the number of active quark flavors, $\alpha_s = 0.3$ is the strong coupling, $h_D = \Gamma(\frac{8}{3})\zeta(\frac{5}{3})(2\pi)^{2/3} \simeq 1.81$ is a numerical prefactor, q_D is the Debye wavenumber defined by $q_D^2 = g^2 N_f \mu_B^2/(18\pi^2)$, and $g^2 = 4\pi\alpha_s$. In our numerical calculations we set $\alpha_s = 0.3$ [MGA Can we find some standard citation about the running of the QCD coupling that would show that this is an appropriate value at energy scales of order 400 MeV?][35, 45] .

In (C1) the first term comes from the interaction mediated by Landau-damped transverse gluons, and the second term comes from Debye-screened longitudinal [MGA and transverse?] gluons. The first term is larger at temperatures up to $\sqrt{\alpha_s}\mu_B/6$, which is of order 70 MeV, but because both terms have similar T dependence they are already comparable (within a factor of 2) when the temperature rises above $0.04\sqrt{\alpha_s}\mu_B/3$, which is of order 7 MeV. In neutron star mergers the temperature can range from below 1 MeV (early inspiral) to about 70 MeV, so for our numerical calculations we use the full expression, but for the analytic approximations of Secs. III C and IV B we only use the Landau damping term.

AH shall I add the derivation of K diffusion from individual quark currents that I have typed up in the notes?

Appendix D: Heat Transport

[LB This entire appendix needs clarification.]

The nonadvective thermal current $q^\mu_{\rm th}$, the term that describes heat flowing relative to the fluid, satisfies $U_\mu q^\mu_{\rm th} = 0$ and is purely spatial in that frame [41, 42].[LB isn't q only spatial in the rest frame of the fluid?][JS In the formula of Eckart heat law Eq. (D1), $(g^{\mu\nu} + U^\mu U^\nu)\nabla_\nu T = (g^\mu_{\ \nu} + U^\mu U_\nu)\nabla^\nu T$, the part $g^\mu_{\ \nu} + U^\mu U_\nu$ is a projection operator that projects any 4 vector (in this case the covariant derivative of temperature field $\nabla^\nu T$) to the hypersurface that U^μ is normal to, so if there's no acceleration, meaning $TU^\alpha \partial_\alpha U^\mu = 0$, then $q^\mu_{\rm th}$ should live in the hypersurface whose normal vector is U^μ, i.e. q^μ is spacial in the baryon current rest frame, but not in the phase boundary rest frame.]

For stationary profiles in which the heat current adjusts much faster than the conditions experienced by the fluid change [LB What conditions?], we use the first-order Eckart heat law [41, 43, 52],

$$\begin{aligned} q^\mu_{\rm th} = -\kappa_{\rm th} \big[&(g^{\mu\nu} + U^\mu U^\nu)\nabla_\nu T \\ &+ T U^\alpha \nabla_\alpha U^\mu \big] , \end{aligned} \tag{D1}$$

where $\kappa_{\rm th} \geq 0$ is the thermal conductivity. The first term is the temperature gradient projected into the local fluid rest space; the second is the relativistic acceleration correction.

We impose steady planar flow in flat spacetime, so all fields depend only on x. With $U^\mu = (\gamma, u, 0, 0)$, orthogonality gives $q^0_{\rm th} = u q^x_{\rm th}/\gamma$, and Eq. (D1) gives

$$q^x_{\rm th} = -\kappa_{\rm th} \left[(1+u^2)\frac{dT}{dx} + Tu\frac{du}{dx} \right] . \tag{D2}$$

Keeping the transport terms to leading order in $u \sim 3 \times 10^{-7}$ ($u^2 \sim 10^{-13}$), while retaining the relativistic form of the ideal-fluid fluxes, is consistent with the adiabatic numerical check of relative n_B and P variations below 10^{-10} and 10^{-12}, respectively (Appendix G). This gives Fourier's law,

$$q^x_{\rm th} \simeq -\kappa_{\rm th} \frac{dT}{dx} , \tag{D3}$$

The minus sign directs heat toward lower temperature. In quark matter, $\kappa_{\rm th}$ is determined by quark scattering [44].

We use independent density-gradient Fick and temperature-gradient Fourier laws, neglecting thermodiffusion (the Soret effect) and composition-driven heat

flow (the Dufour effect). The Fick law uses a gradient of n_K rather than of μ_K/T. We also neglect energy carried by K interdiffusion, so the nonadvective energy current in Eq. (5) is approximated by conduction alone. These are additional transport assumptions, not consequences of slow flow, and they are retained in both thermal limits. The acceleration correction is small where $|Tu\,du/dx| \ll |dT/dx|$. The thermal limits below are taken after this slow-flow reduction; exactly constant T does not by itself remove the acceleration term in Eq. (D2).

Appendix E: Shape of the diffusive flux profile

The diffusive flux $g(x) \equiv -D_K(x)a'(x)$ enters I_1 (Eq. (29)) only through its shape as a function of a rather than x: since $a(x)$ is monotonic, we can trade x for a and write $I_1 = \int_0^{a(0^+)} g(a)\,da$, the area under g between the equilibrated downstream value $a = 0$, where $g = 0$, and the interface value $a = a(0^+)$, where the jump condition Eq. (27) fixes $g(a(0^+)) = \lambda_n u(0^-)[1 - a(0^+)]$.

Write $G \equiv g(a(0^+)) = \lambda_n u(0^-)[1 - a(0^+)]$ for this interface value. Three limiting shapes bracket the possibilities for the area under g. If g rises linearly from 0 to G over $0 \le a \le a(0^+)$, the area is a triangle, $I_1 = \frac{1}{2}Ga(0^+)$. If g instead jumps to G immediately and stays there, the area is the full rectangle, $I_1 = Ga(0^+)$. If g instead stays near 0 over almost the whole range and only rises to G at the very end, the area shrinks toward 0. These are the $\xi = 0$, $\xi = 1$, and $\xi = -1$ limits of

$$I_1 = \frac{1+\xi}{2}\,Ga(0^+)\,, \tag{E1}$$

which is Eq. (31) written in terms of G: ξ measures whether the diffusive flux builds up early ($\xi > 0$) or late ($\xi < 0$) across the conversion region, relative to a linear ramp ($\xi = 0$). For each input $T(0^-)$ and $n_B(0^-)$, there is one unique ξ that makes Eq. (31) exact.

The full numerical solutions to Eq. (19) are smooth between the two endpoints, and after comparing the plots of $g(a)$ using profiles that numerically solve Eq. (37), we find that $\xi = -0.5$ is a good approximation, meaning that $g(a)$ is mostly convex. We use $\xi = -0.5$ in Eq. (32) and Eq. (42) throughout.

Appendix F: Numerical method

[MGA If we want to keep this appendix then we should flesh it out with some info about how to solve the equations, e.g. about shooting with trial values of $u(0^-)$ to find the right value.]

We solve Eq. (19) at fixed temperature in the isothermal case, or at fixed energy flux (38) in the adiabatic case. We compactify the semi-infinite quark region using

$$x = -L_0 \ln(1-s)\,, \qquad 0 \le s \le 1 - \epsilon_{\rm tail}\,, \tag{F1}$$

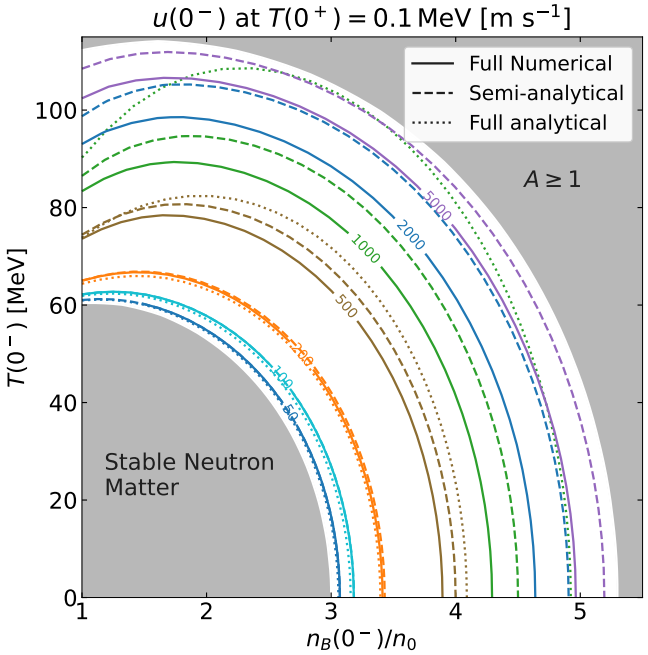


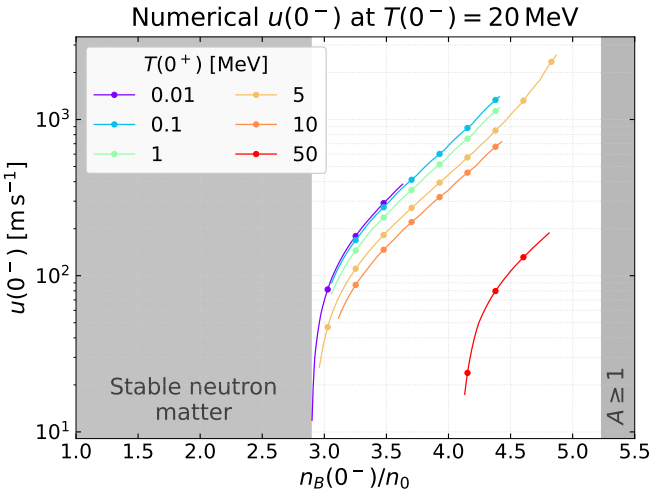
FIG. 7: Comparing the full numerical, semi-analytical, and full analytical methods for $T(0^+) = 0.1$ MeV, with contour levels up to 5000 m/s wherever supported by the calculated data.

where L_0 is fixed from the initial interface diffusion–reaction scale. The result is checked for stability under changes of both L_0 and $\epsilon_{\rm tail}$. We first attempt the requested $a(0^+)$ directly; if necessary, we continue from a smaller interface fraction and use each converged profile as the initial guess for the next step. States with negative temperature, nonpositive density, a failed equation-of-state closure, or nonphysical transport coefficients are rejected. For the isothermal and adiabatic cases, these profiles provide the numerical reference against which the piecewise-constant results in Eqs. (34) and (42), respectively, are tested.

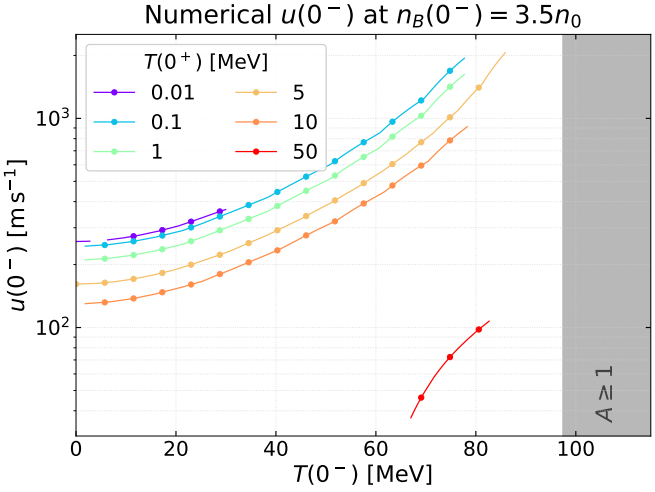
Appendix G: Local thermal equilibrium

[LB We might be able to turn this into a sentence in the main body.]However, $T(0^+) = 0$ lies outside the validity of the present diffusion treatment. As $T(0^+) \to 0$, the quark mean free path $\lambda_Q^{\rm mfp}$ diverges, so quark matter cannot remain in local thermal equilibrium (LTE) across a conversion layer with a finite variation scale. Requiring the mean free path to remain shorter than the scale on which the diffusion coefficient changes,

$$\lambda_Q^{\rm mfp} = 3D_K < \frac{D_K}{|dD_K/dx|}\,, \tag{G1}$$



(a)



(b)

FIG. 8: Dependence of the numerical phase-boundary speed on the prescribed interface temperature $T(0^+)$. Panel (a) shows $u(0^-)$ as a function of $n_B(0^-)$ at fixed $T(0^-) = 20\,\mathrm{MeV}$, while panel (b) shows $u(0^-)$ as a function of $T(0^-)$ at fixed $n_B(0^-) = 3.5n_0$. The velocity axis is logarithmic, the markers denote the numerical slice samples, and the line color runs from colder (purple and blue) to hotter (red) values of $T(0^+)$. The gray regions are excluded in the corresponding slice by stable neutron matter or by $A \geq 1$.

and using the constant-pressure trajectory and interface gradient gives

$$5u(0^-)\lambda_n a(0^+)[1 - a(0^+)] < A^2 - a^2(0^+)\,. \tag{G2}$$

The LTE-limited composition is obtained by saturating this inequality,

$$A^2 - \left[a^{\mathrm{LTE}}(0^+)\right]^2 = 5\lambda_n u(0^-) a^{\mathrm{LTE}}(0^+)\left[1 - a^{\mathrm{LTE}}(0^+)\right]\,, \tag{G3}$$

with $u(0^-)$ evaluated at $a(0^+)$ from Eq. (42).

Finally, the full numerical solutions confirm the scale estimate: across the conversion layer, the relative variations of n_B and P are typically no larger than 10^{-10} and 10^{-12}, respectively. The piecewise-constant approximation removes variation that is numerically invisible while retaining the composition and temperature profiles that set the conversion rate.

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